



UNIVERSITI SAINS ISLAM MALAYSIA
جَامِعَةُ الْعُلُومِ الْإِسْلَامِيَّةِ الْمَالِيزِيَّةِ
ISLAMIC SCIENCE UNIVERSITY OF MALAYSIA

SKJ 3113 INTERNET AND NETWORK SECURITY

SEMESTER A242

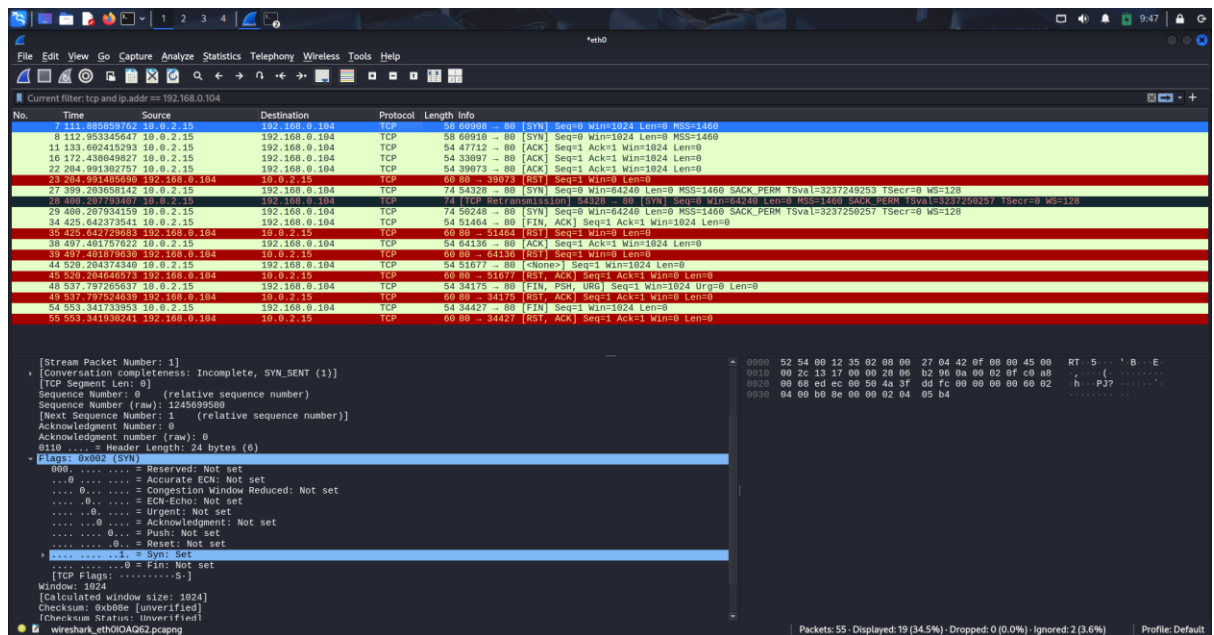
LAB 3 ASSIGNMENT

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MSC 2

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1. Scan summary table



Scan Type	Flags used	Target IP	Hostname	Target response	Type of response received
SYN scan	SYN	192.168.0.104	10.0.2.15	Yes	SYN
Connect scan	SYN+ACK	192.168.0.104	10.0.2.15	Yes	SYN-ACK
ACK scan	ACK	192.168.0.104	10.0.2.15	Yes	RST
Maimon scan	FIN+ACK	192.168.0.104	10.0.2.15	Yes	RST,ACK
Window scan	ACK	192.168.0.104	10.0.2.15	Yes	RST
Null scan	NO FLAG	192.168.0.104	10.0.2.15	Yes	RST,ACK
Xmas scan	FIN, PSH, URG	192.168.0.104	10.0.2.15	Yes	RST,ACK
FIN scan	FIN	192.168.0.104	10.0.2.15	Yes	RST,ACK

2. Analysis of Response

Scan type	Response	Port status	Notes
SYN	SYN	Open	Target responded
Connect	SYN-ACK	Open	TCP handshake completed
ACK	RST	Closed	Helps determine firewall
Maimon	RST,ACK	Closed	Used to detect filtered ports
Window	RST	Closed	Can reveal additional information through window size
Null	RST,ACK	Closed	No flag set, used to bypass simple firewalls
Xmas	RST,ACK	Closed	The targeted port is closed
FIN	RST,ACK	Closed	Common response to FIN scans on closed port

3- Tcp Flags diagram

For SYN and SYN-ACK

```
[Stream Packet Number: 1]
  [Conversation completeness: Incomplete, SYN_SENT (1)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 1245699580
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 0
  Acknowledgment number (raw): 0
  0110 .... = Header Length: 24 bytes (6)
  Flags: 0x002 (SYN)
    000. .... = Reserved: Not set
    ...0 .... = Accurate ECN: Not set
    .... 0... = Congestion Window Reduced: Not set
    .... .0.. = ECN-Echo: Not set
    .... ..0. = Urgent: Not set
    .... ...0 = Acknowledgment: Not set
    .... .... 0... = Push: Not set
    .... ..0.. = Reset: Not set
    .... ...1. = Syn: Set
    .... .... 0 = Fin: Not set
  [TCP Flags: .....S.]
  Window: 1024
  [Calculated window size: 1024]
  Checksum: 0xb08e [unverified]
  [Checksum Status: Unverified]
  wireshark_eth0IOAQ62.pcapng
```

```
[Stream Packet Number: 1]
  [Conversation completeness: Incomplete (4)]
  [TCP Segment Len: 0]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 0
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 3791722701
  0101 .... = Header Length: 20 bytes (5)
  Flags: 0x010 (ACK)
    000. .... = Reserved: Not set
    ...0 .... = Accurate ECN: Not set
    .... 0... = Congestion Window Reduced: Not set
    .... .0.. = ECN-Echo: Not set
    .... ..0. = Urgent: Not set
    .... ...1 = Acknowledgment: Set
    .... .... 0... = Push: Not set
    .... ..0.. = Reset: Not set
    .... .... 0 = Syn: Not set
    .... .... 0 = Fin: Not set
  [TCP Flags: .....A....]
  Window: 1024
  [Calculated window size: 1024]
  [Window size scaling factor: -1 (unknown)]
  wireshark_eth0IOAQ62.pcapng
```

0000	52 54 00 12 35 02 08 00 27 04 42
0010	00 28 d9 57 00 00 33 06 e1 59 0a
0020	00 68 ba 60 00 50 00 00 00 00 e2
0030	04 00 29 36 00 00

Packets: 55 · Displayed: 19 (34.5%) · Dropped: 0 (0.0%) · Ignored: 2 (3.6%) · Profile: Default

For ACK

```
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (4)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 0
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 2771236106
0101 ... = Header Length: 20 bytes (5)
▼ Flags: 0x010 (ACK)
  000. .... = Reserved: Not set
  ...0 .... = Accurate ECN: Not set
  ....0 .... = Congestion Window Reduced: Not set
  ....0 .... = ECN-Echo: Not set
  ....0 .... = Urgent: Not set
  ....1 .... = Acknowledgment: Set
  ....0 .... = Push: Not set
  ....0 .... = Reset: Not set
  ....0 .... = Syn: Not set
  ....0 .... = Fin: Not set
[TCP Flags: .....A....]
Window: 1024
[Calculated window size: 1024]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x02e4 [unverified]
wireshark_eth0IOAQ62.pcapng
```

For Maimon Scan

```
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (52)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 0
[Next Sequence Number: 2 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 906958683
0101 ... = Header Length: 20 bytes (5)
▼ Flags: 0x011 (FIN, ACK)
  000. .... = Reserved: Not set
  ...0 .... = Accurate ECN: Not set
  ....0 .... = Congestion Window Reduced: Not set
  ....0 .... = ECN-Echo: Not set
  ....0 .... = Urgent: Not set
  ....1 .... = Acknowledgment: Set
  ....0 .... = Push: Not set
  ....0 .... = Reset: Not set
  ....0 .... = Syn: Not set
  ....1 .... = Fin: Set
[TCP Flags: .....A...F]
Window: 1024
[Calculated window size: 1024]
[Window size scaling factor: -1 (unknown)]
Checksum: 0xc7f1 [unverified]
wireshark_eth0IOAQ62.pcapng
```

For window scan

```
[Stream Packet Number: 1]
▶ [Conversation completeness: Incomplete (36)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 0
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1667448990
0101 ..... = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
000. .... = Reserved: Not set
...0 .... = Accurate ECN: Not set
.... 0... = Congestion Window Reduced: Not set
.... .0.. = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...1 = Acknowledgment: Set
.... .... = Push: Not set
.... ..0. = Reset: Not set
.... ...0. = Syn: Not set
.... ....0 = Fin: Not set
[TCP Flags: .....A....]
Window: 1024
[Calculated window size: 1024]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x3fdh [unverified]
wireshark_eth0IOAQ62.pcapng
```

For null

```
[Stream Packet Number: 1]
▶ [Conversation completeness: Incomplete (36)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 324004364
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
0101 ..... = Header Length: 20 bytes (5)
Flags: 0x000 (<None>)
000. .... = Reserved: Not set
...0 .... = Accurate ECN: Not set
.... 0... = Congestion Window Reduced: Not set
.... .0.. = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...0 = Acknowledgment: Not set
.... .... = Push: Not set
.... ..0. = Reset: Not set
.... ...0. = Syn: Not set
.... ....0 = Fin: Not set
[TCP Flags: .....]
Window: 1024
[Calculated window size: 1024]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x173c [unverified]
wireshark_eth0IOAQ62.pcapng
```

For Xmas

```
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (36)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 4156379782
[Next Sequence Number: 2 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
0101 .... = Header Length: 20 bytes (5)
▼ Flags: 0x029 (FIN, PSH, URG)
  000. .... = Reserved: Not set
  ...0 .... = Accurate ECN: Not set
  ....0... = Congestion Window Reduced: Not set
  ....0... = ECN-Echo: Not set
  ....1. .... = Urgent: Set
  ....0... = Acknowledgment: Not set
  ....1... = Push: Set
  ....0... = Reset: Not set
  ....0... = Syn: Not set
  ....1 = Fin: Set
  [TCP Flags: .....U.P..F]
Window: 1024
[Calculated window size: 1024]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x0e89 [unverified]
wireshark_eth0IOAQ62.pcapng
```

For FIN

```
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (36)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 3427571011
[Next Sequence Number: 2 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
0101 .... = Header Length: 20 bytes (5)
▼ Flags: 0x001 (FIN)
  000. .... = Reserved: Not set
  ...0 .... = Accurate ECN: Not set
  ....0... = Congestion Window Reduced: Not set
  ....0... = ECN-Echo: Not set
  ....0... = Urgent: Not set
  ....0... = Acknowledgment: Not set
  ....0... = Push: Not set
  ....0... = Reset: Not set
  ....0... = Syn: Not set
  ....1 = Fin: Set
  [TCP Flags: .....F]
Window: 1024
[Calculated window size: 1024]
[Window size scaling factor: -1 (unknown)]
Checksum: 0xf668 [unverified]
wireshark_eth0IOAQ62.pcapng
```

4- Observations and finding

By analyzing TCP header fields across different scan types, we can know port status, firewall rules and OS. SYN scans are the most suitable for port discovery. ACK, FIN, Xmas and Null help bypass firewall and detect filtering whilst Window and Maimon scans provide OS specific insight. RST used to reject new connections and RST commonly used in resetting malformed usually used in Linux.